

Sean Luka Girgis

Senior Python Platform / Data Reliability Engineer | Production Systems, SQL, Airflow & Observability

214-315-2190 | seanlgirgis@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Senior Python, data reliability, and production engineering professional with 20+ years of enterprise experience across Python 3, SQL, Linux/Unix, ETL/data workflows, REST API concepts, Airflow, Docker, CI/CD concepts, monitoring/observability, production troubleshooting, and resilient long-running pipeline design. Strong background modernizing fragile manual processes into repeatable automated workflows, validating production outputs, supporting critical enterprise operations, and building documented systems that are easier to monitor, troubleshoot, and maintain. Best aligned to Python platform/data reliability roles requiring scalable Python workflows, SQL databases, Linux, Airflow, Docker, Git, production monitoring, fault-tolerant processing patterns, and enterprise troubleshooting; Azure, AKS/Kubernetes, Azure Container Apps, Service Bus, Key Vault, Redis, Solr, and deep cloud-native backend ownership are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark and SQL workflows that processed telemetry from 6,000+ infrastructure endpoints into validated dashboards, reports, forecasting datasets, and operational insights for critical banking infrastructure.
- Modernized manual capacity and reporting processes into repeatable Python-driven pipelines with validation checkpoints, source-feed checks, output comparisons, and documented recovery procedures.
- Designed resilient long-running data workflows around extraction, transformation, validation, reconciliation, publishing, monitoring, and troubleshooting of operational reporting datasets.
- Implemented production checks for row counts, missing or late feeds, nulls, duplicate keys, unmatched joins, freshness, abnormal metric movement, and output consistency against prior trusted runs.
- Troubleshot production issues by tracing source feeds, Python processing, SQL transformations, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Used AWS S3, Glue, Athena, Redshift, Oracle, EC2/ECS concepts, PySpark, and SQL to support scalable data processing, warehouse-backed reporting, and cloud-oriented pipeline patterns.
- Created runbooks, support procedures, metric definitions, validation logic, data flow notes, and troubleshooting documentation to improve maintainability and team handoff.
- Partnered with infrastructure, application, analytics, engineering, and business teams to prioritize incidents, explain findings, coordinate fixes, and improve reliability of production reporting workflows.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Provided application performance and production visibility support for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.
- Investigated production and pre-production performance issues by correlating application monitoring data, synthetic transaction signals, and load-test behavior.

- Created dashboards and reporting views that converted monitoring signals into actionable reliability, performance, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical reliability improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services application monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance data, replacing manual reporting workflows with repeatable operational processes.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and production-readiness testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCT/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported phased cutover, troubleshooting, documentation, and defect resolution for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.

- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Maintained code in CVS and ClearCase and supported legacy application environments across HP-UX, SunOS, Linux, WebLogic, WebSphere, Oracle, and messaging interfaces.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Production Data Reliability and RCA Workflow

Built validation and troubleshooting workflows for production reporting datasets, including row-count checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and documented recovery steps.

Technologies: Python, SQL, Pandas, Oracle, Data Validation, RCA, Runbooks

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL, Data Quality

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation